

Experiment/Case Study No : 5

Aim:

Apply array methods and object handling; create a cart total calculator with discount logic.

Software/Tools Required

- Visual Studio Code
- Google Chrome (Web Browser)

Experiment Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Advanced Shopping Cart</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Advanced Shopping Cart</h1>

    <section class="input-group">
      <h2>Add Product</h2>
      <div class="form-row">
        <label>Product Name</label>
        <input type="text" id="pName" placeholder="e.g. Laptop">
      </div>
      <div class="form-row">
        <label>Price (₹)</label>
        <input type="number" id="pPrice" placeholder="e.g. 1200">
      </div>
      <button id="btnAdd">Add to Cart</button>
    </section>

    <section class="cart-display">
      <div class="cart-header">
        <h2>Cart Items</h2>
        <button id="btnClear" class="btn-danger">Clear Cart</button>
      </div>
      <table id="cartTable">
        <thead>
          <tr>
            <th>Product</th>
            <th>Price (₹)</th>
            <th>Action</th>
          </tr>
        </thead>
      </table>
    </section>
  </div>
</body>
</html>
```

```

    </thead>
    <tbody></tbody>
  </table>
</section>

<div id="expensive-section" class="alert-box warning"></div>
<div id="summary-section" class="alert-box success"></div>
</div>
<div style="position: fixed; bottom: 10px; right: 10px; font-family: sans-serif; font-size:
14px; color: #da1111; background-color: rgba(255, 208, 0, 0.8); padding: 5px 10px;
border-radius: 4px; z-index: 1000;">
  Name: M HUSSAIN KHAN <br>
  PRN: 8482871166
</div>

<script src="script.js"></script>
</body>
</html>

```

```

const AppState = {
  cart: [],
  discountRate: 0.10,
  expensiveThreshold: 1000
};

const DOM = {
  nameInput: document.getElementById('pName'),
  priceInput: document.getElementById('pPrice'),
  btnAdd: document.getElementById('btnAdd'),
  btnClear: document.getElementById('btnClear'),
  tableBody: document.querySelector('#cartTable tbody'),
  expensiveSection: document.getElementById('expensive-section'),
  summarySection: document.getElementById('summary-section')
};

function init() {
  DOM.btnAdd.addEventListener('click', handleAddProduct);
  DOM.btnClear.addEventListener('click', handleClearCart);
  DOM.priceInput.addEventListener('keypress', (e) => {
    if (e.key === 'Enter') handleAddProduct();
  });
  render();
}

function handleAddProduct() {
  const name = DOM.nameInput.value.trim();
  const price = parseFloat(DOM.priceInput.value);

  if (!name || isNaN(price) || price <= 0) {
    alert('Please enter a valid product name and a positive price.');
```

```

const newProduct = {
  id: Date.now(),
  name: name,
  price: price
};

AppState.cart.push(newProduct);
DOM.nameInput.value = "";
DOM.priceInput.value = "";
render();
}

function handleRemoveProduct(id) {
  // ARRAY METHOD: filter - removes the item with the matching ID
  AppState.cart = AppState.cart.filter(item => item.id !== id);
  render();
}

function handleClearCart() {
  if (confirm('Are you sure you want to clear the entire cart?')) {
    AppState.cart = [];
    render();
  }
}

function getExpensiveProducts() {
  // ARRAY METHOD: filter - returns only items > 1000
  return AppState.cart.filter(item => item.price > AppState.expensiveThreshold);
}

function calculateTotals() {
  const totalItems = AppState.cart.length;

  // ARRAY METHOD: reduce - sums up all prices
  const totalBeforeDiscount = AppState.cart.reduce((sum, item) => sum + item.price, 0);

  // ARRAY METHOD: some - checks if at least one item is expensive
  const hasExpensiveItem = AppState.cart.some(item => item.price > AppState.expensiveThreshold);

  const discountAmount = totalBeforeDiscount * AppState.discountRate;
  const finalTotal = totalBeforeDiscount - discountAmount;

  return {
    totalItems,
    totalBeforeDiscount,
    discountAmount,
    finalTotal,
    hasExpensiveItem
  };
}

```

```

function renderExpensiveWarning() {
  const expensiveItems = getExpensiveProducts();

  if (expensiveItems.length > 0) {
    // ARRAY METHOD: map - extracts just the names for the message
    const itemNames = expensiveItems.map(item => item.name).join(', ');

    DOM.expensiveSection.innerHTML = `
      <strong> ⚠ High Value Alert:</strong>
      You have ${expensiveItems.length} expensive product(s) (>
      ₹${AppState.expensiveThreshold}):
      <em>${itemNames}</em>.
    `;
    DOM.expensiveSection.style.display = 'block';
  } else {
    DOM.expensiveSection.style.display = 'none';
  }
}

function renderSummary() {
  const stats = calculateTotals();

  DOM.summarySection.innerHTML = `
    <h3>Order Summary</h3>
    <p>Items: ${stats.totalItems}</p>
    <p>Subtotal: ₹${stats.totalBeforeDiscount.toFixed(2)}</p>
    <p>Discount (10%): -₹${stats.discountAmount.toFixed(2)}</p>
    <hr style="border: 0; border-top: 1px solid #86efac; margin: 10px 0;">
    <p><strong>Total Payable: ₹${stats.finalTotal.toFixed(2)}</strong></p>
  `;
  DOM.summarySection.style.display = 'block';
}

function renderTable() {
  DOM.tableBody.innerHTML = "";

  if (AppState.cart.length === 0) {
    DOM.tableBody.innerHTML = '<tr><td colspan="3" style="text-align:center;">Your cart
is empty.</td></tr>';
    return;
  }

  // ARRAY METHOD: forEach - iterates to create HTML rows (side effect)
  AppState.cart.forEach(item => {
    const row = document.createElement('tr');
    row.innerHTML = `
      <td>${item.name}</td>
      <td>₹${item.price.toFixed(2)}</td>
      <td><button class="btn-remove"
onclick="handleRemoveProduct(${item.id})">Remove</button></td>
    `;
    DOM.tableBody.appendChild(row);
  });
}

```

```

});
}

function render() {
  renderTable();
  renderExpensiveWarning();
  renderSummary();
}

document.addEventListener('DOMContentLoaded', init);

:root {
  --primary: #2563eb;
  --success: #16a34a;
  --warning: #ca8a04;
  --danger: #dc2626;
  --bg: #f3f4f6;
  --card: #ffffff;
  --text: #1f2937;
}

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background-color: var(--bg);
  color: var(--text);
  margin: 0;
  padding: 20px;
  line-height: 1.6;
}

.container {
  max-width: 800px;
  margin: 0 auto;
  background: var(--card);
  padding: 2rem;
  border-radius: 12px;
  box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
}

h1, h2 { text-align: center; color: #111827; }

.input-group {
  background: #f8fafc;
  padding: 1.5rem;
  border-radius: 8px;
  margin-bottom: 2rem;
  border: 1px solid #e2e8f0;
}

.form-row { margin-bottom: 1rem; }
label { display: block; font-weight: 600; margin-bottom: 0.5rem; }
input {
  width: 100%;

```

```

padding: 0.75rem;
border: 1px solid #cbd5e1;
border-radius: 6px;
box-sizing: border-box;
font-size: 1rem;
}

button {
width: 100%;
padding: 0.75rem;
background-color: var(--primary);
color: white;
border: none;
border-radius: 6px;
font-weight: 600;
cursor: pointer;
transition: background 0.2s;
}

button:hover { background-color: #1d4ed8; }

.btn-danger {
background-color: var(--danger);
width: auto;
padding: 0.5rem 1rem;
font-size: 0.9rem;
}
.btn-danger:hover { background-color: #b91c1c; }

.cart-header {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 1rem;
}

table {
width: 100%;
border-collapse: collapse;
margin-bottom: 2rem;
}

th, td {
padding: 1rem;
text-align: left;
border-bottom: 1px solid #e5e7eb;
}

th { background-color: #f1f5f9; font-weight: 600; }

.btn-remove {
background-color: #fee2e2;
color: var(--danger);

```

```

padding: 0.4rem 0.8rem;
font-size: 0.85rem;
}
.btn-remove:hover { background-color: #fecaca; }

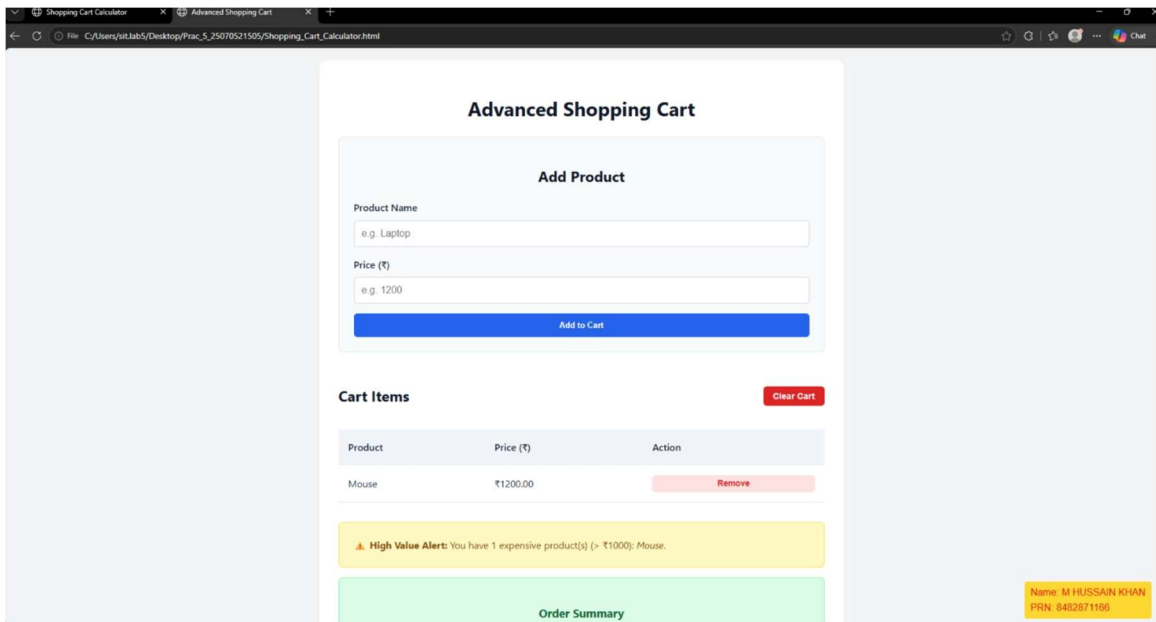
.alert-box {
padding: 1.5rem;
border-radius: 8px;
margin-bottom: 1rem;
display: none;
}

.alert-box.warning {
background-color: #fef9c3;
border: 1px solid #fde047;
color: #854d0e;
}

.alert-box.success {
background-color: #d4edda;
border: 1px solid #c3e6cb;
color: #28a745;
text-align: center;
}

```

Output



FileC:/Users/MohammadHussain%20Khan/Music/Temp/Shopping_Cart_Calculator.html

Add Product

Product Name

e.g. Laptop

Price (₹)

e.g. 1200

Add to Cart

Cart Items

Clear Cart

Product	Price (₹)	Action
Headphones	₹500.00	Remove

Order Summary

Items: 1

Subtotal: ₹500.00

Discount (10%): -₹50.00

Total Payable: ₹450.00

Name: M HUSSAIN KHAN

PRN: 8482971166

Case Study Title

Advance shopping cart with minimum and maximum value of the product

Case Study Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Advanced Shopping Cart</title>
  <link rel="stylesheet" href="Style1.css">
</head>
<body>
  <div class="container">
    <h1>Advanced Shopping Cart</h1>

    <div class="input-section">
      <h3>Add Product</h3>
      <div class="form-group">
        <label>Product Name</label>
        <input type="text" id="productName" placeholder="e.g. Laptop">
      </div>
      <div class="form-group">
        <label>Price (₹)</label>
        <input type="number" id="productPrice" placeholder="e.g. 1200">
      </div>
      <button onclick="addToCart()">Add to Cart</button>
    </div>

    <div class="cart-section">
      <div class="cart-header">
        <h3>Cart Items</h3>
        <button class="clear-btn" onclick="clearCart()">Clear Cart</button>
      </div>
      <table>
        <thead>
          <tr>
            <th>Product</th>
            <th>Price (₹)</th>
            <th>Action</th>
          </tr>
        </thead>
        <tbody id="cartTableBody">
        </tbody>
      </table>
      <p id="emptyMessage">Your cart is empty.</p>
    </div>

    <div class="summary-section">
      <h3>Order Summary</h3>
      <div class="summary-row">
        <span>Items: <span id="itemCount">0</span></span>
      </div>
      <div class="summary-row">
        <span>Subtotal:</span>
        <span>₹<span id="subtotal">0.00</span></span>
      </div>
    </div>
  </div>
</body>
</html>
```

```

    </div>
    <div class="summary-row">
      <span>Discount (10%):</span>
      <span>-₹<span id="discount">0.00</span></span></div>
    <div class="summary-row total">
      <span>Total Payable:</span>
      <span>₹<span id="total">0.00</span></span></div>
    <div class="stats-row">
      <span>Max Value: ₹<span id="maxValue">0.00</span></span></div>
      <span>Min Value: ₹<span id="minValue">0.00</span></span></div>
    </div>
  </div>
  <script src="script1.js"></script>
  <div style="position: fixed; bottom: 10px; right: 10px; font-family: sans-serif;
font-size: 14px; color: #da1111; background-color: rgba(255, 208, 0, 0.8); padding: 5px
10px;
border-radius: 4px; z-index: 1000;">
    Name: M HUSSAIN KHAN <br>
    PRN: 8482871166
  </div>
</body>
</html>

```

```

let cart = [];

function addToCart() {
  const nameInput = document.getElementById('productName');
  const priceInput = document.getElementById('productPrice');

  const name = nameInput.value.trim();
  const price = parseFloat(priceInput.value);

  if (name === '' || isNaN(price) || price <= 0) {
    alert('Please enter a valid product name and price.');
```

return;

```
  }

  cart.push({ name, price });

  nameInput.value = '';
  priceInput.value = '';

  updateUI();
}

function removeFromCart(index) {
  cart.splice(index, 1);
  updateUI();
}

function clearCart() {
  cart = [];
  updateUI();
}

function updateUI() {
  const tbody = document.getElementById('cartTableBody');
  const emptyMsg = document.getElementById('emptyMessage');
  const itemCountSpan = document.getElementById('itemCount');
  const subtotalSpan = document.getElementById('subtotal');
  const discountSpan = document.getElementById('discount');
  const totalSpan = document.getElementById('total');
```

```

const maxSpan = document.getElementById('maxValue');
const minSpan = document.getElementById('minValue');

tbody.innerHTML = '';

if (cart.length === 0) {
  emptyMsg.style.display = 'block';
  maxSpan.textContent = '0.00';
  minSpan.textContent = '0.00';
} else {
  emptyMsg.style.display = 'none';

  const prices = cart.map(item => item.price);
  const maxPrice = Math.max(...prices);
  const minPrice = Math.min(...prices);

  maxSpan.textContent = maxPrice.toFixed(2);
  minSpan.textContent = minPrice.toFixed(2);

  cart.forEach((item, index) => {
    const row = document.createElement('tr');
    row.innerHTML = `
      <td>${item.name}</td>
      <td>₹${item.price.toFixed(2)}</td>
      <td><button class="remove-btn"
onclick="removeFromCart(${index})">Remove</button></td>
    `;
    tbody.appendChild(row);
  });
}

const count = cart.length;
const subtotal = cart.reduce((sum, item) => sum + item.price, 0);
const discount = subtotal * 0.10;
const total = subtotal - discount;

itemCountSpan.textContent = count;
subtotalSpan.textContent = subtotal.toFixed(2);
discountSpan.textContent = discount.toFixed(2);
totalSpan.textContent = total.toFixed(2);
}

@import
url('https://fonts.googleapis.com/css2?family=Fredoka+One&family=Nunito:wght@400;700&display=swap');

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: 'Nunito', sans-serif;
}

body {
  background-color: #E0F7FA;
  background-image: radial-gradient(#4DD0E1 15%, transparent 16%), radial-
gradient(#4DD0E1 15%, transparent 16%);
  background-size: 60px 60px;
  background-position: 0 0, 30px 30px;
  display: flex;
  justify-content: center;
  padding: 20px;
  min-height: 100vh;
}

```

```

.container {
  background-color: #ffffff;
  width: 100%;
  max-width: 500px;
  border-radius: 25px;
  box-shadow: 0 10px 0 #B2EBF2, 0 10px 20px rgba(0,0,0,0.15);
  overflow: hidden;
  border: 4px solid #4DD0E1;
}

h1 {
  text-align: center;
  padding: 25px;
  color: #FF6F00;
  font-family: 'Fredoka One', cursive;
  font-size: 32px;
  text-shadow: 2px 2px 0px #FFD54F;
  border-bottom: 3px dashed #FFD54F;
  letter-spacing: 1px;
}

h3 {
  font-size: 20px;
  color: #0288D1;
  margin-bottom: 15px;
  font-weight: 700;
}

.input-section {
  background-color: #FFF3E0;
  padding: 25px;
  border-bottom: 3px dashed #FFCC80;
}

.form-group {
  margin-bottom: 15px;
}

.form-group label {
  display: block;
  margin-bottom: 8px;
  font-size: 16px;
  color: #E65100;
  font-weight: bold;
}

input {
  width: 100%;
  padding: 12px;
  border: 2px solid #FFCC80;
  border-radius: 15px;
  font-size: 16px;
  background-color: #fff;
  transition: all 0.3s;
  outline: none;
}

input:focus {
  border-color: #FF6F00;
  box-shadow: 0 0 8px rgba(255, 111, 0, 0.3);
  transform: scale(1.02);
}

button {
  width: 100%;
  padding: 12px;
}

```

```

        background-color: #FF6F00;
        color: white;
        border: none;
        border-radius: 50px;
        cursor: pointer;
        font-size: 18px;
        font-weight: bold;
        font-family: 'Fredoka One', cursive;
        box-shadow: 0 5px 0 #E65100;
        transition: all 0.2s;
        text-transform: uppercase;
    }

    button:hover {
        background-color: #FF8F00;
        transform: translateY(-2px);
        box-shadow: 0 7px 0 #E65100;
    }

    button:active {
        transform: translateY(2px);
        box-shadow: 0 2px 0 #E65100;
    }

    .cart-section {
        padding: 25px;
        background-image: linear-gradient(#fff 50%, #FFFDE7 50%);
        background-size: 100% 40px;
    }

    .cart-header {
        display: flex;
        justify-content: space-between;
        align-items: center;
        margin-bottom: 15px;
    }

    .clear-btn {
        width: auto;
        padding: 8px 15px;
        background-color: #EF5350;
        font-size: 14px;
        border-radius: 20px;
        box-shadow: 0 4px 0 #C62828;
    }

    .clear-btn:hover {
        background-color: #F44336;
        box-shadow: 0 6px 0 #C62828;
        transform: translateY(-1px);
    }

    table {
        width: 100%;
        border-collapse: separate;
        border-spacing: 0 5px;
        margin-bottom: 15px;
    }

    th {
        color: #0277BD;
        font-weight: 800;
        text-transform: uppercase;
        font-size: 14px;
        padding: 10px;
    }

```

```

td {
  background-color: #E1F5FE;
  padding: 12px;
  font-size: 15px;
  color: #333;
  font-weight: 600;
}

td:first-child { border-top-left-radius: 15px; border-bottom-left-radius:
15px; }
td:last-child { border-top-right-radius: 15px; border-bottom-right-radius:
15px; text-align: right; }

.remove-btn {
  background-color: #FFCA28;
  color: #5D4037;
  padding: 6px 12px;
  font-size: 12px;
  width: auto;
  border-radius: 20px;
  box-shadow: 0 3px 0 #FFA000;
  font-family: 'Nunito', sans-serif;
  font-weight: bold;
}

.remove-btn:hover {
  background-color: #FFD54F;
  box-shadow: 0 5px 0 #FFA000;
  transform: translateY(-1px);
}

.emptyMessage {
  text-align: center;
  color: #90A4AE;
  font-style: italic;
  font-weight: bold;
  font-size: 16px;
  padding: 30px 0;
  background: #fff;
  border-radius: 15px;
  border: 2px dashed #CFD8DC;
}

.summary-section {
  background-color: #C8E6C9;
  padding: 25px;
  border-top: 4px dashed #81C784;
  border-radius: 0 0 20px 20px;
}

.summary-row {
  display: flex;
  justify-content: space-between;
  margin-bottom: 12px;
  font-size: 16px;
  color: #1B5E20;
  font-weight: 700;
}

.summary-row.total {
  font-weight: 900;
  font-size: 22px;
  margin-top: 15px;
  border-top: 2px dashed #81C784;
  padding-top: 15px;
}

```

```
    color: #2E7D32;
    background: #fff;
    padding: 15px;
    border-radius: 15px;
    box-shadow: 0 4px 0 #A5D6A7;
}

.stats-row {
    display: flex;
    justify-content: space-between;
    margin-top: 15px;
    font-size: 14px;
    color: #558B2F;
    background: rgba(255,255,255,0.6);
    padding: 10px;
    border-radius: 15px;
    font-weight: bold;
}
```

Output

This page says
Please enter a valid product name and price.

OK

e.g. 1200

ADD TO CART

Cart Items

CLEAR CART

PRODUCT	PRICE (₹)	ACTION
Your cart is empty.		

Order Summary

Items: 0

Subtotal: ₹0.00

Discount (10%): -₹0.00

Total Payable: ₹0.00

Name: M HUSSAIN KHAN
PRN: 8482671166

Add Product

Product Name

e.g. Laptop

Price (₹)

e.g. 1200

ADD TO CART

Cart Items

CLEAR CART

PRODUCT	PRICE (₹)	ACTION
pen	₹10.00	REMOVE
mouse	₹1000.00	REMOVE

Order Summary

Items: 2

Subtotal: ₹1010.00

Discount (10%): -₹101.00

Total Payable: ₹909.00

Max Value: ₹1000.00 Min Value: ₹10.00

Name: M HUSSAIN KHAN
PRN: 8482671166

Result/Conclusion

The program was executed successfully. Array methods and object handling were used to manage product information, calculate the cart subtotal, apply the required discount logic, and display the final cart total correctly.